Reference Variable

A reference variable is an alias, that is, another name for an already existing variable.
Once a reference is initialized with a variable, either the variable name or the reference
name may be used to refer to the variable.

Syntax :
datatype & refname=existingvariable.

Example
int data=10;
int &ref=data ; // ref is a reference to data

example program:

```
#include <iostream>
using namespace std;
int main()
{
// declaring the variables
int number=88;
int &refNumber=number; // refNumber is a reference to
number
cout<<"number:"<<number<<endl;
cout<<"refNumber:"<<refNumber<<endl;
refNumber++;
cout<<"number:"<<number<<endl;
cout<<"refNumber:"<<refNumber<<endl;
}
output:
number:88
refNumber:88
```

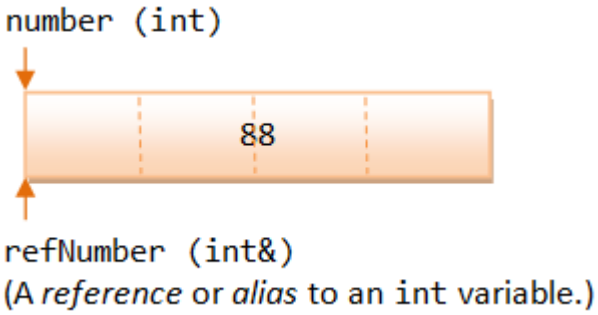
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number:89
refNumber:89

Applications of Reference variable

The main use of references is acting as function formal parameters to support pass-by-reference. In an reference variable is passed into a function, the function works on the original copy (instead of a clone copy in pass-by-value). Changes inside the function are reflected outside the function.

A reference is similar to a pointer. In many cases, a reference can be used as an alternative to pointer, in particular, for the function parameter.



Points to remember:

- A reference variable must be initialized at the time of declaration.
- The reference variable once defined to refer to a variable cannot be changed to point to other variable.
- You cannot create an array of references the way it is possible with pointer. It is mostly used for function argument lists and function return values.

Pointer Vs Reference Variable:

Reference Variable	Pointer
A reference can be said as an alias to an existing variable.	Pointer variables are variables that store address.
Syntax: Data type&newName = existingName;	Syntax: Data type *pntr;

A reference variable cannot be reassigned.
Example :
int x=12;
int &ref=x;
ref is a reference to x as long as it is alive, it cannot rerefered to another variable.
	A pointer can be reassigned.
	Example:
	int x=12;
	int y=15;
	int *ptr;
	ptr=&x;
	ptr=&y // reassigned

A reference variable shares the same address as the original variable.	A pointer has its own memory address and stores it on the stack.
A reference cannot have a null value assigned.	A pointer can have a null value assigned directly.
A reference must be initialized on declaration.	A Pointer may or may not be initialized on declaration.
Dereferencing of reference variable is not required to	Dereferencing of pointer is

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access the data.	required to access the data.
References are not used in DMA	Pointers are used in DMA

Note :

Reference Books : Taken contents and diagrams from various websites.

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